

KILLARNEY, ON, Canada

WMO: 714600

Lat: 45.9682N

Lon: 81.4888W

Elev: 196

StdP: 98.99

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.8	-20.8	-30.9	0.2	-21.9	-27.8	0.3	-18.9	15.5	-4.7	13.5	-3.9	3.2	40	0.624

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.7	25.0	18.1	23.6	17.6	22.4	17.5	20.6	22.7	19.8	21.7	19.1	21.1	5.3	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.8	14.8	21.7	19.0	14.1	21.1	18.2	13.4	20.5	60.2	22.6	57.5	21.8	55.1	21.3	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.6	11.0	9.5	DB	-27.3	28.5	3.1	2.0	-29.5	29.9	-31.4	31.1	-33.1	32.2	-35.4	33.7	(4)
(5)			WB	-27.6	21.9	3.0	1.6	-29.8	23.0	-31.6	23.9	-33.3	24.8	-35.5	25.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.9	-8.8	-7.9	-3.2	3.5	10.3	16.0	18.9	18.7	15.2	8.7	2.2	-4.2	(6)	
	DBStd	10.80	6.95	5.86	5.33	3.96	3.46	3.10	2.48	2.40	3.44	4.01	4.71	5.81	(7)	
	HDD10.0	2480	583	502	409	196	37	1	0	0	4	74	233	440	(8)	
	HDD18.3	4648	841	735	668	444	249	81	23	24	103	300	483	698	(9)	
	CDD10.0	972	0	0	0	2	48	182	277	271	159	33	1	0	(10)	
	CDD18.3	98	0	0	0	0	1	12	41	36	8	0	0	0	(11)	
	CDH23.3	151	0	0	0	0	2	26	74	46	3	0	0	0	(12)	
	CDH26.7	14	0	0	0	0	0	3	7	5	0	0	0	0	(13)	
(14) Wind	WSAvg	4.2	4.5	4.5	4.2	4.2	3.8	3.5	3.6	3.6	4.0	4.9	5.1	4.9	(14)	
(15) Precipitation	PrecAvg	991	73	54	57	82	90	84	84	94	100	109	88	77	(15)	
	PrecMax	1171	120	112	139	147	156	141	137	168	192	177	138	155	(16)	
	PrecMin	811	38	20	8	31	33	29	38	47	38	48	41	33	(17)	
	PrecStd	97	21	24	28	29	31	28	28	28	35	35	25	24	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.2	5.3	9.7	15.5	22.1	26.2	27.5	26.9	23.7	18.6	12.1	8.2	(19)	
		MCWB	4.3	3.6	4.5	7.7	14.1	16.7	19.7	20.1	18.9	15.5	9.6	6.6	(20)	
	2%	DB	3.5	3.0	6.7	12.5	18.8	23.5	25.2	24.5	21.9	17.0	10.6	6.1	(21)	
		MCWB	2.4	1.6	2.8	6.3	13.0	16.3	18.4	18.5	18.4	14.4	8.8	4.7	(22)	
	5%	DB	2.1	1.5	4.7	10.4	16.7	21.6	23.8	23.1	20.8	15.8	9.5	4.6	(23)	
		MCWB	1.0	0.5	1.7	5.4	11.7	15.8	17.5	18.0	17.8	13.5	7.9	3.4	(24)	
	10%	DB	0.8	0.4	3.1	8.5	15.1	20.1	22.5	22.1	19.8	14.6	8.5	3.1	(25)	
		MCWB	-0.2	-0.6	0.8	4.5	10.8	15.6	17.4	17.9	17.2	12.3	6.9	2.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.7	4.0	6.6	9.7	16.7	19.8	21.7	21.8	20.7	16.6	10.5	7.1	(27)	
		MCDB	4.9	4.7	7.6	12.9	19.7	22.4	25.1	24.7	22.1	18.0	11.5	7.9	(28)	
	2%	WB	2.6	1.9	3.7	7.7	14.5	18.6	20.5	20.6	19.6	15.3	9.2	5.0	(29)	
		MCDB	3.2	2.5	5.6	10.4	17.0	20.9	23.1	22.6	21.0	16.5	10.2	5.8	(30)	
	5%	WB	1.2	0.7	2.2	6.3	13.1	17.7	19.7	19.9	18.7	13.9	8.3	3.5	(31)	
		MCDB	2.0	1.4	3.9	9.2	15.4	19.9	22.0	21.6	20.1	15.4	9.4	4.5	(32)	
	10%	WB	-0.1	-0.5	1.2	5.1	11.7	16.8	18.9	19.1	17.8	12.7	7.1	2.1	(33)	
		MCDB	0.8	0.4	2.8	7.9	14.3	19.2	21.3	21.2	19.5	14.4	8.3	3.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.6	7.8	7.1	6.3	6.6	6.0	5.7	5.3	5.5	5.2	5.3	6.1	(35)	
		MCDBR	6.1	6.8	6.9	8.7	8.6	7.8	7.1	6.4	5.8	5.5	5.6	5.8	(36)	
	5% WB	MCWBR	5.9	6.5	5.0	5.2	4.6	4.1	3.8	4.0	4.3	4.8	5.6	5.7	(37)	
		MCWBR	6.0	6.7	6.1	7.3	7.2	6.1	5.9	5.0	5.0	5.0	5.4	5.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.284	0.288	0.309	0.349	0.366	0.396	0.402	0.392	0.364	0.343	0.311	0.276	(40)	
		taud	2.371	2.323	2.380	2.361	2.405	2.340	2.355	2.376	2.439	2.469	2.505	2.471	(41)	
	Ebn at Noon		821	891	922	909	901	871	861	856	852	819	789	800	(42)	
		Edh at Noon	73	93	102	115	114	122	119	112	97	81	65	60	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.21	2.15	3.59	4.61	5.47	5.91	5.99	5.06	3.88	2.30	1.25	0.89	(44)	
	RadStd		0.16	0.22	0.33	0.40	0.51	0.45	0.33	0.30	0.38	0.27	0.17	0.08	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page